Mario Rafael Silva

Assistant Professor of Economics

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Education

PhD Economics, University of California, Irvine Sep 2012 - June 2017

- Dissertation: "Essays on Liquidity, Monopolistic Competition, and Search Frictions"
- Committee: Guillaume Rocheteau, William Branch, Fabio Milani

BS Mathematics, Stanford University Sep 2005 - Jun 2009

Academic and Professional Experience

Assistant Professor August 2021 - present

Department of Accountancy, Economics, and Finance, Hong Kong Baptist University

Assistant Professor Oct 2017 - July 2021

Department of Economics and Finance, Tongji University

UC Irvine Teaching Assistant September 2012 - June 2017

Basic and intermediate micro-and macroeconomics, intermediate econometrics, game theory, global economy

Data Analyst Mar 2012 - Aug 2012

ECON ONE RESEARCH, Boutique Economic Consulting Firm

Processed (cleaned, standardized, grouped, sorted, etc.) transaction level data for multiple firms for an alleged horizontal price fixing conspiracy in the polyfoam industry that settled for $433 million (primarily using SAS)

Research assistant June 2013 - August 2013

Project with Professors Linda Cohen and Amihai Glazer linking alumni university donations to patents

- Conducted extensive research and analysis on the correlation between alumni university donations and patents, under the guidance of Professors Linda Cohen and Amihai Glazer (Bloomberg Law, Westlaw Campus)

Published and Working Papers

Published

- **Silva, Mario R.** and Urias, Marshall. "Productive Demand and Sectoral Capacity Utilization," *Economic Modelling*, 2025
  - While the Solow residual is widely used as a measure of technological progress, it also captures fluctuations in input utilization and other factors. *We develop a multisector model to decompose capacity utilization and attribute variation in the Solow residual to demand, technology, and mismeasurement. Bayesian estimation with sectoral utilization data shows search demand shocks explain most variance in output and utilization, and replicate observed sectoral comovement. Impulse response analysis further demonstrates that demand shocks uniquely generate three-way comovement among utilization rates and the Solow residual.*
- **Silva, Mario R.** and Gabrovski, Miroslav. "Unemployment and Labor Productivity Comovement: The Role of Firm Exit." *Journal of Economic Dynamics and Control*, 2025.
  - *We extend the Diamond-Mortensen-Pissarides model to incorporate sunk entry costs and distinct business destruction/match separation channels, resolving empirical discrepancies in unemployment-productivity comovement and matching observed dynamic labor market correlations.*
- **Silva, Mario R.** "Unsecured Credit, Product Variety, and Unemployment Dynamics." *Macroeconomic Dynamics*, 2020.
  - *Analyzes feedback between revolving credit and product development using an endogenous borrowing constraint and firm entry model. Financial shocks are shown to be essential in matching observed unemployment and credit dynamics during the Great Recession.*
- **Silva, Mario R.** "Corporate Finance, Monetary Policy, and Aggregate Demand." *Journal of Economic Dynamics and Control*, 2019.
  - Analyzes how heterogeneity of financial frictions and monopolistic competition influence the pass through of the nominal interest rate to the real lending rate, its transmission into investment, and corporate cash holdings. Firms finance stochastic investment opportunities with either bank-issued credit or money. In line with empirical evidence, the cash-to-sales ratio increases with the extent of financial constraints, and rises with the intensity of competition for financially constrained firms. Financial constraints raise firms' sensitivity to monetary policy; and a mean-preserving spread of financial frictions reduces investment and

output, strengthens transmission, and reduces the external share of finance. Estimates industry-level markups using production method.

- **Silva, Mario R.** "New Monetarism with Endogenous Product Variety and Monopolistic Competition." *Journal of Economic Dynamics and Control*, 2017.
  - *Examines endogenous variety in a New Monetarist framework, showing how inflation interacts with market structure to affect welfare and firm size under CES and variable elasticity preferences. Under variable elasticity of demand, inflation can increase firm size, reduce markups, and raise welfare, even though output is lower. Under CES preferences, the Friedman rule is optimal, and the welfare cost of inflation is high; moreover, this cost increases monotonically with the markup and is higher with endogenous variety than with a fixed product space.*
- Bajaj, Ayushi, Rocheteau, Guillaume, Hu, Tai-Wei, and **Silva, Mario R.** "Decentralizing Constrained-Efficient Allocations in the Lagos-Wright Pure Currency Economy." *Journal of Economic Theory*, 2016.

Working Papers

- Qiu, Zhesheng, **Silva, Mario R.,** and Urias, Marshall. "Role of Endogenous Business Formation and Exit for Unemployment and Vacancy Dynamics."
- Branch, William and **Silva, Mario R**. "Liquidity, Unemployment, and the Stock Market."

Teaching Experience

Hong Kong Baptist University — August 2021 - present

- Data Analytics for Business Decision Making (BUSI2045), Undergraduate, Spring 2025, 2026
  - Python programming-based introduction to data management and basic statistical inference with the aim of developing research design for business. Questionnaire design and hypothesis testing.
- Advanced Macroeconomics (ECON 7800), MSc, 2021-2023
  - Emphasizes key concepts and measurement issues in macroeconomics, computation of equilibrium, the application of overlapping generations models to liquidity and growth, real business cycle models, financial frictions, the role of endogenous firm entry, a liquidity-based model of monetary policy, heterogeneous agents, unemployment, and demand-induced fluctuations.
  - https://github.com/msilva913/ECON-7800-Programs (Course programs for ECON 7800, Advanced Macroeconomics)
- Digital Economy (ECON 4035, ECON 7640), Undergraduate/MSc, Spring 2022, 2025,
  - Focuses on ongoing work on Central Bank Digital Currency, especially as it relates to financial inclusion and liquidity, effects on bank deposit funding and investment, and government financial transfers. Use pilot projects of CBDC in China, the European Union, and other countries as case studies.

Tongji University — February 2018 - July 2021

- Macroeconomics, PhD, Spring 2019, 2020)
- Money and Banking, MSc, Fall 2018, 2019, 2020
  - Advanced undergraduate. Treatment of commodity money, fiat money, inflation, international monetary systems, price surprises, capital, liquidity and financial intermediation, fully backed central bank money, the payments system, bank risk, and financial multipliers.
- Market Structure, Innovation, and the Macroeconomy, MSc, Spring 2018, 2019
  - Masters level course which incorporates insights from industrial organization and imperfect competition and explores their aggregate implications for innovation, growth, trade, and the formation of new goods.

Grants, Honors, and Awards

Research grants (External)

- General Research Fund (GRF)
  - Project Number 12502123, Exercise Year 2023/2024. Bayesian estimation of liquidity-augmented model with endogenous business formation: application to inflation. Project Fund: $454,136 (HKD)
- National Natural Science Foundation of China
  - Research Fund for International Young Scientists (awarded November 2018)

Honors and Awards

- Graduate Dean's Dissertation Fellowship (awarded for Fall 2017)
- Economics Merit Fellowship (2012-2017)
- Summer Research Fellowship (2014, 2015, 2016)
- Pass with Distinction on Macroeconomics Qualifying Exam (June 2013)

Conference Presentations and Seminars

2025| 3rd Asia Pacific Search and Matching Workshop, CUHK; Society for Economic Measurement Conference, Athens (July); Seminar at National University of Kaohsiung (May)

2024| Seminar at the University of Macao (September); Society for Economic Measurement Conference, Atlanta (August); Hong Kong Junior Macroeconomics Workshop (May); Money and Macroeconomics Workshop, Melbourne (March)

2023| Western Economics Association Economics International, Melbourne (April)

2022| HKBU Junior Macro Group (August); National Taiwan University (April)

2021| Seminar at HKBU; Seminar at Universita di Roma Tor Vergata; Seminar at Federal Deposit Insurance Corporation

2019| Peking University HSBC, Shenzen (Oct); Western Economics Association International, San Francisco (July); Seminar at Shanghai University of Finance and Economics (May)

2018| Society of Economic Measurement Conference, Xiamen University (May); Seminar at Ashoka University (April)

2017| Society for Computational Economics Conference, New York (June); Midwest Macroeconomics, Baton Rouge (May)

2016| Western Economics Association International, Portland (June); West Coast Search and Matching Workshop, San Francisco Federal Reserve (May); Midwest Macroeconomics Conference, Purdue University (May)

2015| Midwest Macroeconomics Workshop, St. Louis (May)

Professional Associations

- American Economics Association; Econometric Society; Western Economics Association International; European Economics Association
- Hong Kong Junior Macro Group

Refereeing

- Economic Inquiry
- Macroeconomic Dynamics

Technical skills

- Languages & Software: Python, Julia, Matlab, R, Stata, SAS, Excel, LaTeX, Dynare, Git
- Quantitative methods:, Bayesian estimation (Metropolis-Hastings), panel data, structural VAR, GMM, GLM

Languages

Spanish (fluent), Italian (conversational)